

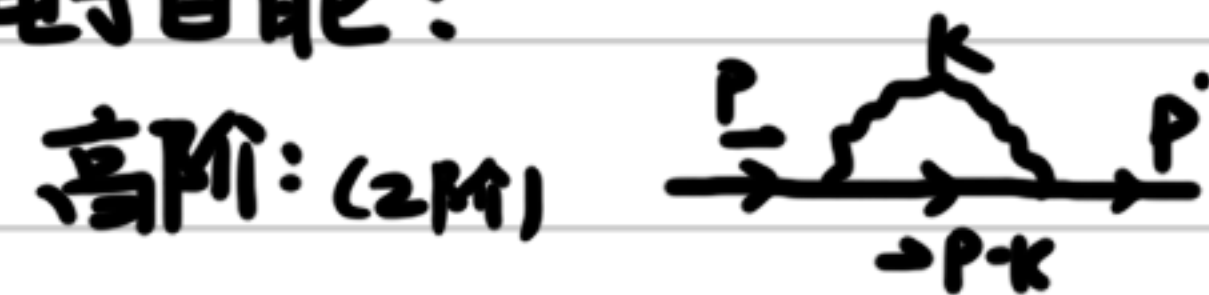
八. 重正化

"将无穷大埋在地毯中" $\longrightarrow$ 量子场论是低能有效理论

1. 发散困难 (自能发散, 顶角发散)

自能积分与正规顶角名称来由: 见 8.5

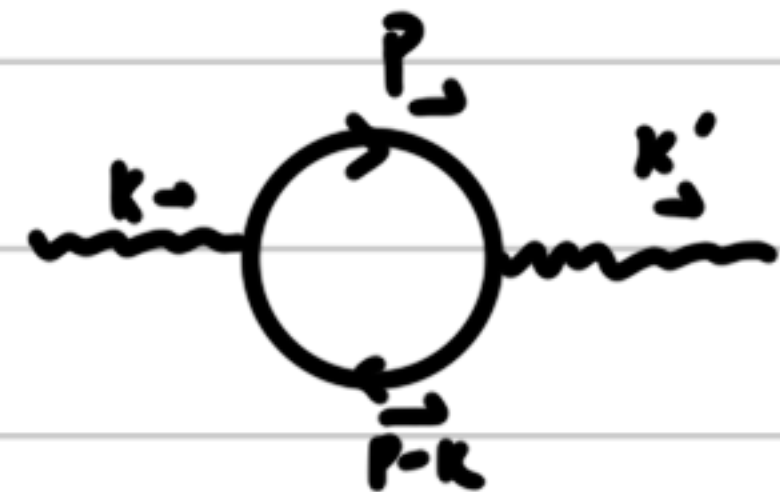
电子自能:



$D = d - 3$



光子自能:



$$\int dx dy dx' dy' i \bar{\eta}_x i S_F(x-y) i \eta_y i \bar{\eta}_{x'} i S_F(x'-y') i \eta_{y'}$$

$D = d - 2$

自能积分 $-i \Sigma^{(2)}(p) = \int \frac{d^d k}{(2\pi)^d} (-ie \gamma^\mu) i D_{F\mu\nu}(k) i S_F(p-k) (-ie \gamma^\nu)$

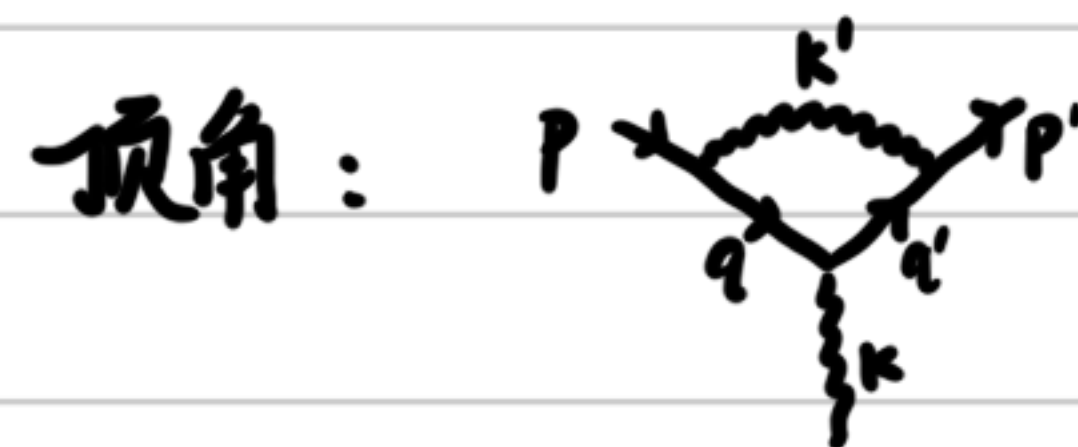
$$= e^2 \int \frac{d^d k}{(2\pi)^d} \frac{-g_{\mu\nu}}{k^2 + i\epsilon} \gamma^\mu \frac{1}{p-k-m+i\epsilon} \gamma^\nu = e^2 \int \frac{d^d k}{(2\pi)^d} F(p, k)$$

$k \rightarrow \infty$ 处发散

自能积分 $-i \Pi^{(2)}(k) = - \int \frac{d^d p}{(2\pi)^d} \text{tr} [(-ie \gamma_\mu) i S_F(p) (-ie \gamma_\mu) i S_F(p-k)]$

$$= -e^2 \int \frac{d^d p}{(2\pi)^d} \text{tr} [\gamma_\mu \frac{p+m}{p^2-m^2} \gamma_\nu \frac{p-k+m}{(p-k)^2-m^2}]$$

$p \rightarrow \infty$ 发散



$D = d - 4$

顶角修正: $\Gamma^{(2)\mu}(p, p') = (2\pi)^d \delta(p'-p-k) \int \frac{d^d k}{(2\pi)^d} (-ie \gamma^\lambda) i S_F(q) (-ie \gamma^\mu) i S_F(q) (-ie \gamma^\nu) i D_{F\mu\nu}(k)$

$$= -ie (2\pi)^d \delta(p'-p-k) (ie \int \frac{d^d k}{(2\pi)^d} \gamma^\lambda \frac{p'-k+m}{(p'-k)^2-m^2} \cdot \gamma^\mu \frac{p-k+m}{(p-k)^2-m^2} \gamma^\nu \frac{1}{k^2} g_{\mu\nu})$$

$$= -ie (2\pi)^d \delta(p'-p-k) (ie \int \frac{d^d k}{(2\pi)^d} \cdot \frac{1}{k^2} \gamma_\lambda \equiv \gamma_\mu \equiv \gamma^\lambda)$$

$\Lambda^{(2)\mu}(p, p')$

无图: $-ie (2\pi)^d \delta(p'-p) \gamma_\mu$

$\hookrightarrow$ 角修正 $\Gamma_\mu^{(2)}(p, p) = \Lambda_\mu^{(2)}(p, p) - \gamma_\mu$

表观发散度:

发散的Feynmann图叫发散图 $\xrightarrow{\text{分解}}$ 原始发散图

表观发散度: 发散维数 $D = \text{维} - \text{分母}$, $D < 0$ 不散

微扰QED: 对圈: 圈: 积分 $D+ = d$
 (加点进行微扰/保证更高阶的图也可极重正化) 光子线 -2, 电子线 -1 } $D = d \cdot l - 2b - f < 0$
 光子 电子

欧拉图: $l = f + b - n + 1$ $\rightarrow D = (d-1)f + (d-2)b - d(n-1)$
 圈 边 顶点

γ -N 顶点: 加外线 $2n = F + 2f$ $\rightarrow D = d + \frac{1}{2}(d-4)n - \frac{1}{2}(d-1)F - \frac{1}{2}(d-2)B$
 $n = B + 2b$ 外线

 $n=4$ 时 $D = 4 - \frac{3}{2}F - B$ 与 n 无关

2. 正规化

分离有效值与无穷大 (一般把无穷写成极限形式, 然后运算)

0. 截断正规化: $\frac{1}{k^2} \rightarrow \frac{1}{k^2} - \frac{1}{k^2 - M^2}$ $M \rightarrow \infty$ (Pauli-Villars 正规化, 虚构一个重矢量场)
 此时 k^2 很大时 M^2 忽略, 为 0

1. 维数正规化:

数学工具

量纲分析:

0.577, Euler 常数
 $\Gamma(x) = \frac{1}{x} - \gamma + O(x)$

作用量 $S = \int d^d x \mathcal{L}$, $[S] = 1 \rightarrow [\mathcal{L}] = L^d = \Lambda^d$
 $\rightarrow \mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \rightarrow [A_\mu] = ([\mathcal{L}] L^2)^{\frac{1}{2}} = \Lambda^{\frac{d}{2}-1}$
 $\rightarrow \mathcal{L} = \bar{\psi}(i \not{\partial} - m)\psi \rightarrow [\psi] = \Lambda^{\frac{d-1}{2}}$
 $\rightarrow \mathcal{L} = -q \bar{\psi} i \not{A} \psi \rightarrow [q] = \Lambda^{2-\frac{d}{2}}$

Feynmann 折叠公式 ($\frac{1}{ab} = \frac{1}{b-a} (\frac{1}{a} - \frac{1}{b}) = \frac{1}{a} \int_0^1 \frac{dx}{x^2}$)

(a, b 可复, 避奇点) $\frac{1}{ab} = \int_0^1 \frac{dx}{[ax+b(1-x)]^2} \xrightarrow{x \rightarrow 0} \frac{1}{ab} = \int_0^1 dx dy \delta(x+y-1) \frac{n y^{n-1}}{(ax+by)^{n+1}}$

$\rightarrow$ 归并 $\frac{1}{a_1 a_2 a_3 \dots} = \int_0^1 dx_1 dx_2 \dots dx_n \delta(\sum x_i - 1) \frac{(n-1)!}{(a_1 x_1 + a_2 x_2 + \dots)^n}$

$\rightarrow \frac{1}{a_1^m a_2^{n_1} \dots} = \frac{\Gamma(m_1 + \dots + m_n)}{\Gamma(m_1) \Gamma(m_2) \dots} \int_0^1 dx_1 dx_2 \dots \delta(\sum x_i - 1) \frac{\pi^{\frac{n-1}{2}}}{(\sum a_i x_i)^{\sum m_i}} \quad (m_i \text{ 可非整数})$

d 维的 γ 矩阵 $\gamma^0 \dots \gamma^{d-1}$

沿用定义: $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$, $\delta^\mu_\mu = d = \gamma^\mu \gamma_\mu$

$$\hookrightarrow \gamma^\mu \gamma^\nu \gamma_\mu = (-d+2)\gamma^\nu$$

加入 $f(d)$, 有: $f(4)=4$, 目的: $\square$

$$\text{tr}(\text{奇数个 } \gamma) = 0, \text{tr}(1) = \underline{f(d)}, \text{tr}(\gamma^\mu \gamma^\nu) = \underline{f(d)} g^{\mu\nu}$$

d 维动量积分

$$P(1.3) \quad I(m^2, \alpha) = \int \frac{d^d p}{(2\pi)^d} \frac{1}{(p^2 - m^2)^\alpha} = \frac{(-1)^{\alpha-1} i}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma(\alpha - \frac{d}{2})}{\Gamma(\alpha)} \frac{1}{m^{2\alpha-d}} \quad (\text{用折叠公式})$$

$\downarrow$ Wick 转动

$$O(4) \quad I(m^2, \alpha) = (-1)^\alpha i \int \frac{d^d p_E}{(2\pi)^d} \frac{1}{(p_E^2 + m^2)^\alpha} \quad (\text{结果同上}) \quad (\text{用球坐标})$$

$\downarrow$ 平移

$$\int d^d p F(p) = \int d^d p F(p+q) \quad I(m^2+q^2, \alpha) = \int \frac{d^d p}{(2\pi)^d} \frac{1}{(p^2 + 2pq + m^2)^\alpha}$$

$\downarrow$ 对 q^μ 求导

$$\int \frac{d^d p}{(2\pi)^d} \frac{p^\mu}{(p^2 + 2pq + m^2)^\alpha} = \frac{\partial I(m^2+q^2, \alpha)}{\partial q_\mu} = \frac{(-1)^{\alpha-1} i}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma(\alpha - \frac{d}{2})}{\Gamma(\alpha)} \frac{q^\mu}{(q^2 + m^2)^{\alpha-\frac{d}{2}}}$$

$$\int \frac{d^d p}{(2\pi)^d} \frac{p^\mu p^\nu}{(p^2 + 2pq + m^2)^\alpha} = \frac{\partial^2 I}{\partial q_\mu \partial q_\nu} = \frac{(-1)^{\alpha-1} i g^{\mu\nu}}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma(\alpha - \frac{d}{2})}{2\Gamma(\alpha)} \frac{1}{(m^2)^{\alpha-1-\frac{d}{2}}}$$

微扰正规化

耦合常数

$$\text{电子自能} \quad \Sigma^{(1)}(p) = i\mu^{4-d} e^2 \int \frac{d^d k}{(2\pi)^d} \frac{-g_{\mu\nu}}{k^2} \gamma^\mu \frac{1}{\not{p} - \not{k} - m} \gamma^\nu = i\mu^{4-d} e^2 \int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2} \frac{(d-2)(\not{p} - \not{k}) - md}{(p-k)^2 - m^2} \stackrel{\text{费曼}}{=} -i\mu^{4-d} e^2 \int \frac{d^d k}{(2\pi)^d} \frac{1}{(k+p)^2} \frac{(d-2)k + md}{k^2 - m^2}$$

重整化把发散转移到了

计算量上!

$$\text{折叠公式} \rightarrow = -i\mu^{4-d} e^2 \int_0^1 dx \int \frac{d^d k}{(2\pi)^d} \frac{(d-2)k + md}{[(p^2 + 2pk + m^2)x + (k^2 - m^2)]} \\ = \frac{-i\mu^{4-d} e^2 (-1)^d}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma(2-d/2)}{\Gamma(2)} \int_0^1 \frac{dx [-(d-2)p^2 x + md]}{[p^2 x^2 + m^2 - (p^2 + m^2)x]^{2-\frac{d}{2}}} \stackrel{x \rightarrow 1-x}{=} \mu^{4-d} e^2 \frac{\Gamma(2-\frac{d}{2})}{(4\pi)^{\frac{d}{2}}} \int_0^1 \frac{dx [- (d-2)p^2(1-x) + md]}{[m^2 x - p^2 x(1-x)]^{2-\frac{d}{2}}}$$

令 $d=4-\epsilon$

$$\Sigma^{(1)}(p) = \frac{e^2}{4\pi} \Gamma(\frac{\epsilon}{2}) \int_0^1 \frac{dx [- (2-\epsilon)p^2(1-x) + md]}{[m^2 x - p^2 x(1-x)]^{\frac{\epsilon}{2}}} \approx \frac{e^2}{4\pi} \Gamma(\frac{\epsilon}{2}) \int_0^1 dx [(-2p^2(1-x) + 4m) + \epsilon(p^2(1-x) - m)] - \frac{\epsilon}{2} (-2p^2(1-x) + 4m) \ln \frac{m^2 x - p^2 x(1-x)}{4\pi\mu^2}$$

$$\frac{e^2}{4\pi} \left[\left(\frac{2}{\epsilon} - \gamma \right) \int_0^1 dx [-2p^2(1-x) + 4m] + 2 \int_0^1 dx p^2(1-x) - \int_0^1 dx (-2p^2(1-x) + 4m) \ln \frac{m^2 x - p^2 x(1-x)}{4\pi\mu^2} \right]$$

$$= \frac{e^2}{4\pi} \left[\left(\frac{2}{\epsilon} - \gamma + \ln 4\pi + \frac{1}{2} - \ln \frac{m^2}{\mu^2} \right) (4m + p^2) + \frac{e^2}{4\pi} \int_0^1 dx (2px - 4m) \ln \left(1 - \frac{p^2}{m^2} x \right) \right]$$

光子自能

方法: 1. 列自能积分 (d维)

2. Feynmann 折叠变分图

3. $I(m^2, m)$ 积分 (专注于发散项)

4. $d=4-\epsilon$, 对 ϵ 小量展开

$$\Pi_{\mu\nu}^{(2)}(k) = i\mu^{4-d}e^2 \int \frac{d^d P}{(2\pi)^d} \text{tr} \left[\gamma_\mu \frac{\not{P}+m}{P^2-m^2} \gamma_\nu \frac{\not{P}-\not{k}+m}{(P-k)^2-m^2} \right] \frac{f(d)}{(P^2-m^2)(P-k)^2-m^2} [2P_\mu P_\nu - P_\mu k_\nu - P_\nu k_\mu - g_{\mu\nu}(P^2 - P \cdot k - m^2)]$$

Feynmann 折叠: 分母: $[(P-k)^2-m^2]x + (P^2-m^2)(1-x) = [(P-kx)^2-m^2+k^2x(1-x)]^2$ $P \rightarrow P+kx$

积分项 $\int \frac{d^d P}{(2\pi)^d} \frac{2P_\mu P_\nu - P_\mu k_\nu - P_\nu k_\mu - g_{\mu\nu}(P^2 - P \cdot k - m^2)}{[(P-kx)^2-m^2+k^2x(1-x)]^2}$ $\xrightarrow{P \rightarrow P+kx}$ $\int \frac{d^d P}{(2\pi)^d} \frac{2P_\mu P_\nu - 2k_\mu k_\nu x(1-x) - g_{\mu\nu}[P^2 - k^2 x(1-x) - m^2]}{(P-kx)^2-m^2+k^2x(1-x)}^2$
 P -次项为 0

$$\Pi_{\mu\nu}^{(2)}(k) = i\mu^{4-d}e^2 f(d) \int_0^1 dx \frac{-i}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma(1-\frac{d}{2})}{2\Gamma(2)[m^2-k^2x(1-x)]^{1-\frac{d}{2}}} \cdot \left[g_{\mu\nu}(2-d) + \frac{2k_\mu k_\nu x(1-x) - g_{\mu\nu}[k^2x(1-x)+m^2]}{m^2-k^2x(1-x)} \cdot 2(1-\frac{d}{2}) \right]$$

$$= \frac{i\mu^{4-d}e^2 f(d) \Gamma(2-\frac{d}{2})}{(4\pi)^{\frac{d}{2}}} \int_0^1 \frac{-2(g_{\mu\nu}k^2 - k_\mu k_\nu)x(1-x)}{[m^2-k^2x(1-x)]^{2-\frac{d}{2}}} dx$$

$d=4-\epsilon$, $f(d) \approx f(4)=4$

$\pi_{\mu\nu}^{(2)}(k) = \frac{\alpha}{32\pi} (k_\mu k_\nu - g_{\mu\nu} k^2) \left[\frac{2}{\epsilon} - \gamma + \ln 4\pi - \ln \frac{m^2}{\mu^2} - 6 \int_0^1 dx x(1-x) \ln \left[1 - \frac{k^2}{m^2} x(1-x) \right] + O(\epsilon) \right]$ ↑ 发散项

顶角发散

$$\Lambda_{\mu}^{(2)}(P', P) = -i\mu^{4-d}e^2 \int \frac{d^d k'}{(2\pi)^d} \frac{1}{k'^2} \gamma_\nu \frac{\not{P}'-\not{k}'+m}{(P'-k')^2-m^2} \gamma_\mu \frac{\not{P}-\not{k}'+m}{(P-k')^2-m^2} \gamma^\nu$$

Feynmann 折叠: 3个分母: $\frac{1}{abc} = 2 \int_0^1 dx \int_0^{1-x} dy \frac{1}{[ax+by+c(1-x-y)]^3}$

平移: $k' \rightarrow k' - Px - P'y$

分母 $\rightarrow 2 \int_0^1 dx \int_0^{1-x} dy \frac{1}{[(k-Px-P'y)^2-M^2]^3}$ ↑ 提出 k' $M^2 = m^2(x+y) + 2PP'xy - P^2x(1-x) - P'^2y(1-y)$ ↑ k' -次项无贡献: 消去

$$\Lambda_{\mu}^{(2)} = -i2\mu^{4-d}e^2 \int \frac{d^d k'}{(2\pi)^d} \int_0^1 dx \int_0^{1-x} dy \frac{G_{\mu}}{(k^2-M^2)^3}$$

$\downarrow$
 $\Lambda_{\mu}^{(2)} = -i2\mu^{4-d}e^2 \int_0^1 dx \int_0^{1-x} dy \int \frac{d^d k}{(2\pi)^d} \frac{P_{\mu}(P, P', x, y)}{(k^2-M^2)^3} \rightarrow 1/2$

$$\Lambda_{\mu}^{(2)} = -i2\mu^{4-d}e^2 \int_0^1 dx \int_0^{1-x} dy \frac{(d-2)(\gamma_\mu k^2 - k_\mu k^2)}{(k^2-M^2)^3} = \frac{2\mu^{4-d}e^2}{(4\pi)^{\frac{d}{2}}} \int_0^1 dx \int_0^{1-x} dy \frac{\gamma_\mu (d-2)^2 \Gamma(2-\frac{d}{2})}{2\Gamma(3)(M^2)^{2-\frac{d}{2}}}$$

$$\approx \frac{\alpha}{4\pi} \left(\frac{2}{\epsilon} - \gamma + \ln 4\pi \mu^2 - 1 - 2 \int_0^1 dx \int_0^{1-x} dy \ln M^2 \right)$$

↑ 发散项

$G_{\mu} = \gamma_\nu [-\not{P}x + \not{P}'(1-y) - \not{k} + m] \gamma_\mu [\not{P}(1-x) - \not{P}y - \not{k}' + m] \gamma^\nu = K_{\mu}(k) + P_{\mu}(P, P', x, y)$
 $= (d-2)(\gamma_\mu k^2 - 2k_\mu k^2) + \gamma_\nu (-\not{P}x + \not{P}'(1-y) + m) \gamma_\mu (-\not{P}y + \not{P}'(1-x) + m) \gamma^\nu$

3. QED 单圈重正化

renormalization

不刮骨的疗法 → 改变模型

1. 抵消项

QED 拉氏密度 $\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi - e\bar{\psi}\gamma^\mu \psi A_\mu - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}(\partial_\mu A^\mu)^2$

微扰元素: 电子线, 光子线, 顶角

发散项: $\frac{1}{2\pi\epsilon}(4m-p)$ $\frac{2\alpha}{3\pi\epsilon}(k_\mu k_\nu - g_{\mu\nu}k^2)$ $\frac{\alpha\gamma_\mu}{2\pi\epsilon}$

引入抵消项: $\delta\mathcal{L} = \delta Z_2 \bar{\psi}i\gamma^\mu \partial_\mu \psi - \delta m \bar{\psi}\psi - \delta e \bar{\psi}\gamma^\mu \psi A_\mu - \frac{1}{4}\delta Z_3 (\partial_\mu A^\mu - \partial_\nu A^\nu)^2$

待定常数, 使发散项为0

Feynman 规则:

电子二线顶点:  贡献 $i(\delta Z_2 p - \delta m)$

光子二线顶点:  $-i\delta Z_3(g^{\mu\nu}k^2 - k^\mu k^\nu)$

修正三线顶点:  $-i\delta e(2\pi)^4 \delta(\Sigma p_i)\gamma^\mu$

常数确定: 顶点贡献与自能项抵消

电子: $-i\Sigma^{(1)}(p) + i(\delta Z_2 p - \delta m) = -i\Sigma_F^{(1)}(p) \rightarrow \delta Z_2 = \frac{-\alpha}{2\pi\epsilon}, \delta m = \frac{-2\alpha}{\pi\epsilon}m$

光子: $i\Pi_{\mu\nu}^{(1)}(k) - i\delta Z_3(g^{\mu\nu}k^2 - k^\mu k^\nu) = i\Pi_{\mu\nu}^{(1)}(k) \rightarrow \delta Z_3 = \frac{-2\alpha}{3\pi\epsilon}$

顶角: $e\Lambda_\mu^{(1)}(p,p) + \delta e\gamma_\mu = e\Lambda_{F,\mu}^{(1)}(p,p) \rightarrow \delta e = \frac{\alpha}{2\pi\epsilon}e$

最小减除方案 MS

(若去掉发散 + $\gamma\ln 4\pi \rightarrow \overline{MS}$)

重正化: $\mathcal{L} = Z_2 \bar{\psi}i\gamma^\mu \partial_\mu \psi - Z_m m \bar{\psi}\psi - Z_1 e \bar{\psi}\gamma^\mu \psi A_\mu - \frac{1}{4}Z_3 F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}(\partial_\mu A^\mu)^2$

定义 $\psi_B = (Z_2)^{\frac{1}{2}}\psi, A_B^\mu = (Z_3)^{\frac{1}{2}}A^\mu$

$m_B = \frac{Z_m}{Z_2}m, e_B = \sqrt{\frac{Z_1}{Z_2 Z_3}}e, \lambda_B = \frac{\lambda}{Z_3}$

$$\mathcal{L}_B = \bar{\psi}_B i \gamma_\mu \partial_\mu \psi_B - m \bar{\psi}_B \psi_B - e_0 \bar{\psi}_B i \gamma_\mu \psi_B A_\mu^A - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} (\partial_\mu A_\mu^A)^2$$

物理诠释:

bare 裸粒子, 裸量 修正(辐射修正) 抵消项

(高阶微扰的粒子外有粒子包围 $\rightarrow m$ 与 e 与 裸不同)

包圈性无限归一化 $\longrightarrow \psi$ 与 A_μ 与实际观测不同

其它:

$$1. Z_1 = Z_2 \rightarrow e_B = Z_1^{-\frac{1}{2}} e \quad (\text{后证明与阶数无关, 是真空极化效应})$$

$$2. e = \mu^{\frac{\epsilon}{2}} Z_1^{-\frac{1}{2}} e_B = \mu^{\frac{\epsilon}{2}} \left(1 - \frac{e^2}{12\pi^2}\right)^{\frac{1}{2}} e_B \xrightarrow{\epsilon \rightarrow 0} \mu \frac{\partial e}{\partial \mu} = \frac{e^3}{12\pi^2} \rightarrow e^2(\mu) = e^2(\mu_0) \left[1 - \frac{e^2(\mu_0)}{6\pi^2} \ln \frac{\mu}{\mu_0}\right]$$

$\mu = \mu_0 e^{\epsilon \gamma_e} e^{\gamma_e(\mu_0)}$ 为朗道极点

4. 电子反常磁矩

1. 顶角函数

$$\text{Diagram: } \text{fermion line with a loop} = \text{fermion line} + \text{fermion line with a loop} + \dots \rightarrow M_{fi} = -e^2 [\bar{u}(p') \Gamma^\mu(p', p) u(p)] \cdot \frac{1}{q^2} [\bar{u}(k') \gamma^\mu u(k)]$$

↓ 等效外场 $A_\mu^{\text{ext}}(q) = \int \frac{e}{q^2} [\bar{u}(k') \gamma_\mu u(k)]$

$$M_{fi} = e [\bar{u}(p') \Gamma^\mu(p', p) u(p)] A_\mu^{\text{ext}}(q)$$

规范变换: $A_\mu(q) \rightarrow A_\mu(q) + q_\mu \chi(q) \rightarrow M_{fi}$ 不变: $q_\mu \Gamma^\mu(p', p)$ 约束 (规范?)

Gordon 恒等式

$$\text{-1阶 } \Gamma^\mu = \gamma^\mu, \text{ 定义 } \sigma^{\mu\nu} = \frac{i}{2} [\gamma^\mu, \gamma^\nu]$$

$$\gamma^\mu u(p) = \frac{1}{m} (p^\mu - i \sigma^{\mu\nu} p_\nu) u(p)$$

$$\bar{u}(p') \gamma^\mu = \frac{1}{m} \bar{u}(p') (p'^\mu + i \sigma^{\mu\nu} p'_\nu)$$

$$\left\{ \begin{array}{l} \{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} \\ \gamma^\mu p_\mu u(p) = m u(p), \quad \bar{u}(p') \gamma^\mu p'_\mu = \bar{u}(p') m \end{array} \right.$$

$$\text{故 } \bar{u}(p') \gamma^\mu u(p) = \frac{1}{2} \bar{u}(p') (\gamma^\mu u(p)) + \frac{1}{2} (\bar{u}(p') \gamma^\mu) u(p)$$

$$\rightarrow \bar{u}(p') \gamma^\mu u(p) = \frac{1}{2m} \bar{u}(p') [(p'^\mu + p^\mu) + i \sigma^{\mu\nu} q_\nu] u(p), \quad q = p' - p$$

形状因子 (简单顶点的空间分布 $F_1(q^2), F_2(q^2)$ 等)

Gordon 恒等式 $\rightarrow p' + p$ 可表示为 γ 与 $p' - p$

$$\Gamma(p+p, p'-p) \xrightarrow{\quad} \Gamma(\gamma, q)$$

$$\Gamma(p, p) = \gamma^\mu F_1(q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{2m} F_2(q^2)$$

↑ 因为对 $q^\mu, \bar{u}\gamma^\mu q_\mu u = 0$

$$\mathcal{L}_1 = -e\bar{\psi}(x)\gamma^\mu\psi(x)A_\mu(x) \rightarrow \mathcal{L}_1^{\text{eff}} = e\int dy \bar{\psi}(x)\Gamma(x-y)\psi(x)A_\mu(y) \quad , \quad \Gamma(x) = \int \frac{d^4q}{(2\pi)^4} \Gamma(q) e^{-iqx}$$

$$\text{代入上式: } \Gamma(x) = \gamma^\mu F_1(x) - \frac{1}{2m}\sigma^{\mu\nu}\partial_\nu F_2(x) \dots$$

做修正的等效相互作用 $\rightarrow$ 空间中简单顶点的相互作用

$$F_1(x) = \int \frac{d^4q}{(2\pi)^4} F_1(q^2) e^{-iqx}$$

↓ 可测得

$$\text{Mott 散射: } \frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4p^2 v^2 \sin^2(\frac{\theta}{2})} F_1^2(q^2) [1 - v^2 \sin^2(\frac{\theta}{2})]$$

2. 电子磁矩

低阶: Dirac 理论:

$$\text{Gordon 恒等式: } \bar{\psi}\gamma^\mu\psi = \frac{i}{2m}\bar{\psi}[(\vec{\partial}^2 - \vec{\partial}^2) - i\sigma^{\mu\nu}(\partial_\nu + \partial_\nu)]\psi$$

$$\mathcal{L}_{1m} = -\frac{e}{2m}\bar{\psi}\sigma^{\mu\nu}(\partial_\nu + \partial_\nu)\psi A_\mu \quad j^\mu$$

$$\mathcal{L}_{1e} = \frac{ie}{2m}\bar{\psi}\vec{\partial}^2\psi A_\mu = -ej^\mu A_\mu$$

$$\text{经典场 } A_\mu = (V, \vec{A}) \quad \text{正则方程: } \frac{d\vec{p}}{dt} = -\nabla H = e[-\nabla V + \nabla(\vec{v}\cdot\vec{A})] = e[-\nabla V + \nabla\times(\nabla\times\vec{A}) + (\vec{v}\cdot\nabla)\vec{A}]$$

$$\text{Lorentz 力: } \frac{d(\vec{p} + e\vec{A})}{dt} = e(\vec{E} + \vec{v}\times\vec{B})$$

$$\mathcal{L}_{1m} = -\frac{e}{2m}\bar{\psi}\sigma^{\mu\nu}(\partial_\nu + \partial_\nu)\psi A_\mu = \frac{e}{2m}\bar{\psi}\sigma^{\mu\nu}\psi\partial_\nu A_\mu = -\frac{e}{4m}\bar{\psi}\sigma^{\mu\nu}\psi F_{\mu\nu}$$

$$S^i = -i\varepsilon^{ijk}\gamma^j\gamma^k/4 \quad = -2 \frac{e}{2m}\bar{\psi}\vec{S}\psi\vec{B} \rightarrow \mu_0 = \frac{e}{2m} \text{ 磁矩}$$

高阶: 反常磁矩

Pauli 项?

$$\text{顶角修正: } \mathcal{L}_1^{\text{eff}} \approx -\lim_{q\rightarrow 0} e\bar{\psi}(x)\Gamma(q)\psi(x)A_\mu(x) \quad (\text{缓变场近似})$$

$$= \lim_{q\rightarrow 0} [-e\bar{\psi}(x)\gamma^\mu\psi(x)A_\mu(x)F_1(q^2) - \frac{e}{4m}\bar{\psi}(x)\sigma^{\mu\nu}\psi(x)F_{\mu\nu}F_2(q^2)]$$

$$= \lim_{q\rightarrow 0} (F_1(q^2)\mathcal{L}_{1e} + F_2(q^2)\mathcal{L}_{1m})$$

对 Landé g 因子: $g = \frac{1}{2} [F_1(q) + F_2(q)] = 2 [F_1(0) + F_2(0)] \rightarrow g_0 = 2$

$F_1(0) = 1 \rightarrow$ Lorentz 力形式不变 $\rightarrow$

$F_2(0) \rightarrow a = \frac{1}{2}(g - g_0) = F_2(0)$

Schwinger 项

顶角函数: $\Gamma_\mu^{(2)}(P', P) = \gamma_\mu + \Lambda_{K\mu}^{(2)}(P', P) + \Lambda_{P\mu}^{(2)}(P', P)$

壳约束: $P^2 = P'^2 = m^2, (P - P')^2 = q^2 = 0$ (慢变, 场动量 $q \rightarrow 0$)

取 $M^2 = m^2(x+y) + 2m^2xy - m^2(1-x) - m^2(1-y) = m^2(x+y)^2$

$$\int_0^1 dx \int_0^{1-x} dy \ln M = \int_0^1 dx \int_0^{1-x} dy (\ln m^2 + 2 \ln(x+y)) = \frac{1}{2}(\ln m^2 - 1)$$

顶角函数: $\Lambda_{K\mu}^{(2)}(P, P') \approx \frac{\alpha \gamma_\mu}{4\pi} \frac{2}{\epsilon} + \frac{\alpha \gamma_\mu}{4\pi} (-\gamma + \ln 4\pi \frac{\mu^2}{m^2}) \rightarrow F_1(0)$ 贡献

Dirac 方程, P 与 P' 对称

P_μ 项: $P_\mu(P, P', x, y) = 2m^2 \gamma_\mu [(x+y-2)^2 - 2] + 4m [(y-xy-x^2)P_\mu + (x+xy-y^2)P'_\mu]$

$F_1(0)$ 贡献

$F_1(0)$ 与 $F_2(0)$ 贡献:

$$-\frac{\alpha}{4\pi} \int_0^1 dx \int_0^{1-x} dy \frac{4m [(y-xy-x^2)P_\mu + (x+xy-y^2)P'_\mu]}{m^2(x+y)^2} = -\frac{\alpha}{2\pi} \frac{P_\mu + P'_\mu}{2m}$$

QED 的电子磁矩形式为: $\mu_e = \mu_0 [1 + C_1 \frac{\alpha}{\pi} + \dots C_i (\frac{\alpha}{\pi})^i]$

Dirac 项 $\rightarrow$ Schwinger 项 $C_1 = \frac{1}{2} \dots$

Schwinger 项

后面太难算, 略 (用计算机)

5. Ward-Takahashi 恒等式 $\rightarrow$ 证明 QED 的可重整性

1. 自能算符 (又叫质量算符)

$$\text{fermion line} = \text{self-energy diagrams} + \dots$$

传播子去外线: 自能算符 $-i\Sigma(P) = -i\Sigma^{(2)}(P) - i\Sigma^{(4)}(P) - i\Sigma^{(6)}(P) + \dots$

完全传播子: 2点 Green 函数

$$\bullet = \text{---} + \text{---} \circ \text{---} + \text{---} \circ \text{---} \circ \text{---} + \dots$$

$$G(P) = G_0(P) + G_0(P)[-i\Sigma(P)]G_0(P) + \dots$$

$$= G_0(P) [1 + [-i\Sigma(P)G_0(P)] + [-i\Sigma(P)G_0(P)]^2 + \dots]$$

$$= G_0(P) [1 - i\Sigma(P)G_0(P)]^{-1}, \quad \text{标量场 } G_0(P) = \frac{i}{P^2 - m^2 + i\epsilon}$$

标量场:

$$= \frac{i}{P^2 - [m^2 + \Sigma(P)] + i\epsilon} = \frac{i}{P^2 - m_c^2 + i\epsilon}$$

对旋量场, 同上, $G_0 = \frac{i}{P - m + i\epsilon} \rightarrow G(P) = \frac{i}{P - [m + \Sigma_P] + i\epsilon}$

2. 正规顶角

连通 Green 函数生成元: $W[J] \rightarrow$ Legendre 变换: $\Gamma[\phi] = W[J] - \int dx J(x)\phi(x)$

$$\phi(x) = \frac{\delta W[J]}{\delta J(x)}, \quad \phi_c|_{J=0} = \langle 0|\phi(x)|0\rangle = \phi_c(\text{真空})$$

$$\frac{\delta \Gamma[\phi]}{\delta \phi(x)} \Big|_{\phi=\phi_c} = 0 \leftarrow \text{1点函数 } \frac{\delta \Gamma}{\delta \phi(x)} = \int dy \frac{\delta W[J]}{\delta J(y)} \frac{\delta J(y)}{\delta \phi(x)} - \int dy \frac{\delta J(y)}{\delta \phi(x)} \underbrace{\phi(y)}_{\frac{\delta W[J]}{\delta J(y)}} - J(x) = -J(x)$$

$$-i\Gamma = \frac{\delta^2 \Gamma[\phi]}{\delta J(x)\delta J(y)} \Big|_{\phi=\phi_c} = [G_c(x,y)]^{-1} \leftarrow \text{2点函数 } \frac{\delta^2 \Gamma}{\delta \phi_x \delta \phi_y} = -\frac{\delta J(x)}{\delta \phi(y)}$$

$$(G_c'(x,y) = \frac{i\delta^2 W[J]}{\delta J(x)\delta J(y)} = \frac{\delta \phi(x)}{\delta J(y)}) \rightarrow J=0 \text{ 为 } G_c(x,y)$$

$$-i\Gamma G_c' = \int dz (-i\Gamma(x,z)G_c'(z,y)) = \delta(x-y) \rightarrow -i\Gamma \text{ 与 } G_c' \text{ 互逆}$$



3点函数

$$\text{由前: } \int dz \frac{\delta^2 \Gamma[\phi]}{\delta \phi(y)\delta \phi(u)} \frac{\delta^2 W[J]}{\delta J(z)\delta J(w)} = -\delta(y-w)$$

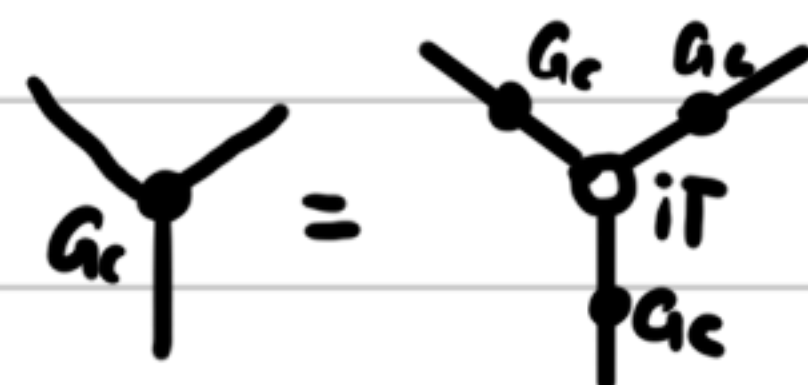
↓ 对 J(u) 变分

$$-\frac{\delta(y-w)}{\delta J(u)} = 0 = \int dz \frac{\delta^2 \Gamma[\phi]}{\delta \phi(y)\delta \phi(z)} \frac{\delta^2 W[J]}{\delta J(u,z,w)} + \int dx dz \frac{\delta \phi(x)}{\delta J(u)} \frac{\delta^2 \Gamma[\phi]}{\delta \phi(x,y,z)} \frac{\delta^2 W[J]}{\delta J(z,w)}$$

$$\int dz \Gamma_\phi(y,z) G_c'(u,z,w) = -\int dx dz i G_c'(x,u) \Gamma(x,y,z) i G_c'(z,w)$$

J=0, 互逆 ↓

$$G_c(u,v,w) = \int dx dy dz G_c(y,v) G_c(x,u) i \Gamma(x,y,z) G_c(z,w)$$



3. Ward-Takahashi恒等式

QED Green 函数: $Z[J, \bar{\eta}, \eta] = N \int DA^\mu D\bar{\psi} D\psi e^{i \int dx (\mathcal{L} + J_\mu A^\mu + \bar{\eta}\psi + \bar{\psi}\eta)}$

定域规范小变换: $A_\mu \rightarrow A_\mu + \partial_\mu \chi$, $\psi \rightarrow e^{ie\chi}\psi$, $\bar{\psi} \rightarrow \bar{\psi} e^{ie\chi}$, $\chi \ll 1$

$$\delta Z = N \int DA^\mu D\bar{\psi} D\psi F_\chi[A^\mu, \bar{\psi}, \psi] e^{i \int dx J_\mu A^\mu + \dots}$$

$$= F_\chi \left[\frac{\delta}{i \delta J_\mu}, \frac{\delta}{i \delta \bar{\eta}}, \frac{\delta}{i \delta \eta} \right] Z[J, \bar{\eta}, \eta]$$

$$F_\chi = i \int dx \delta \left[-\frac{1}{2} (\partial_\mu A^\mu)^2 + J_\mu A^\mu + \bar{\eta}\psi + \bar{\psi}\eta \right]$$

$$= i \int dx \left[-\lambda (\partial_\mu A^\mu) \partial^\mu \chi + J_\mu \partial^\mu \chi + ie\chi (-\bar{\eta}\psi + \bar{\psi}\eta) \right]$$

$$= i \int dx \left[-\lambda \square (\partial_\mu A^\mu) - \partial^\mu J_\mu + ie(\bar{\psi}\eta - \bar{\eta}\psi) \right] \chi$$

$$\delta Z = 0, \text{ 代 } \lambda Z = e^{iW} \rightarrow -\lambda \square \left(\partial_\mu \frac{\delta W}{\delta J_\mu} \right) - \partial^\mu J_\mu - ie \left(\bar{\eta} \frac{\delta W}{\delta \bar{\eta}} - \eta \frac{\delta W}{\delta \eta} \right) = 0$$

对正规顶角: $\Gamma[A^\mu, \bar{\psi}, \psi] = W[J_\mu, \bar{\eta}, \eta] - \int dx (J_\mu A^\mu + \bar{\eta}\psi + \bar{\psi}\eta)$

是否可与辛结构关联? $\star$

$$\begin{cases} \frac{\delta \Gamma}{\delta A^\mu(x)} = -J_\mu & \frac{\delta \Gamma}{\delta \bar{\psi}(x)} = -\eta(x) & \frac{\delta \Gamma}{\delta \psi(x)} = \bar{\eta}(x) \\ \frac{\delta W}{\delta J_\mu} = A^\mu & \frac{\delta W}{\delta \eta} = -\bar{\psi}(x) & \frac{\delta \Gamma}{\delta \bar{\eta}} = -\psi(x) \end{cases}$$

物理含义: 光子位置变对电子效果

等于电子传播与吸收顺序差

坐标: $\chi_x(\dots \overset{x}{\circ} \dots) = ie \left(\overset{x}{\circ} \text{---} \overset{y}{\circ} \right) - ie \left(\text{---} \overset{y}{\circ} \text{---} \overset{x}{\circ} \right)$

动量: $k^\mu(\dots \overset{k}{\circ} \dots) = \left(\overset{k}{\circ} \text{---} \overset{k}{\circ} \right) - \left(\text{---} \overset{k}{\circ} \text{---} \overset{k}{\circ} \right)$

W-T恒等式: $\partial^\mu \frac{\delta^2}{\delta \bar{\psi}(y) \delta \psi(z)}$

$A_\mu = \bar{\psi} = \psi = 0 \rightarrow$

$$i\lambda \square \partial_\mu A^\mu - i\partial^\mu \frac{\delta \Gamma}{\delta A^\mu(x)} - e\bar{\psi} \frac{\delta \Gamma}{\delta \bar{\psi}(x)} + e\psi \frac{\delta \Gamma}{\delta \psi(x)} = 0$$

$$i\partial_x^\mu \frac{\delta^3 \Gamma}{\delta A^\mu(x) \delta \bar{\psi}(y) \delta \psi(z)} \Big|_0 = e\delta(x-y) \frac{-\delta^2 \Gamma}{\delta \psi(z) \delta \bar{\psi}(x)} \Big|_0 + e\delta(x-z) \frac{-\delta^2 \Gamma}{\delta \bar{\psi}(y) \delta \psi(x)} \Big|_0$$

即: 对顶角函数 $\Gamma_\mu(x; y, z)$:

$$\Gamma_\mu(x; y, z) = \frac{-\delta^3 \Gamma}{\delta A^\mu(x) \delta \bar{\psi}(y) \delta \psi(z)}, \quad \Gamma(y, z) = \frac{-\delta^2 \Gamma}{\delta \bar{\psi}(y) \delta \psi(z)}$$

$$i\partial_x^\mu \Gamma_\mu(x; y, z) = e\delta(x-z) \Gamma(y, x) - e\delta(x-y) \Gamma(x, z) = e\delta(x-z) \Gamma(y, z) - e\delta(x-y) \Gamma(y, z) \rightarrow \text{W-T恒等式}$$

$G = i\delta_F \rightarrow -i\Gamma(y, z) = [i\delta_F(y, z)]^{-1}$ δ_F : 完全传播子, 加修正后的传播子

重正化中的关系

$$S_F(x-y) = -i \langle 0 | T \psi(x) \bar{\psi}(y) | 0 \rangle$$

$$\downarrow$$

$$\psi' = Z_2^{1/2} \psi \quad \Gamma^\mu = Z_1^{-1} \gamma^\mu$$

$$S_F = Z_2 S_F \quad \downarrow \quad \text{顶点}$$

$$\text{边} \quad Z_1^{-1} \gamma^\mu = Z_2^{-1} \gamma^\mu \rightarrow Z_1 = Z_2$$

定规不变: $\frac{\partial}{\partial x_\mu} + ieA^\mu \rightarrow \frac{\partial}{\partial x_\mu} + ieB A^\mu$

$eA^\mu = eB A^\mu, eB = Z_1 Z_2^{-1} Z_3^{-1} e, A_B^\mu = Z_3^{-1} A^\mu$

可同得 $Z_1 = Z_2$

$$T(y, z) = T(y, -z) = \int \frac{d^4 p}{(2\pi)^4} S_F(p) e^{-ip(y-z)}$$

$$\downarrow$$

顶角 $\int dx dy dz i T_\mu(x; y, z) e^{i(-kx + py - pz)} = -ie(2\pi)^4 \delta(p' - p - k) T_\mu(k; p', p)$

有: $k^\mu T_\mu(k; p+k, p) = S_F^{-1}(p+k) - S_F^{-1}(p) \rightarrow \text{动量的W-T恒等式}$

$\rightarrow$ 2.5与2.5 Green函数的结果

Ward恒等式: $k^\mu \rightarrow 0$ (红外) $T_\mu(0; p, p) = \frac{\partial S_F^{-1}(p)}{\partial p^\mu}$

形式变换: $G_0 = i S_F(p) = \frac{i}{\not{p} - m + i\epsilon} \rightarrow G = i S_F(p)$

$$S_F^{-1}(p) = \not{p} - m \rightarrow \tilde{S}_F^{-1}(p) = \not{p} - m - \Sigma(p)$$

$$\frac{\partial S_F^{-1}(p)}{\partial p^\mu} = \gamma_\mu \xrightarrow{\text{修正}} \frac{\partial \tilde{S}_F^{-1}(p)}{\partial p^\mu} = \gamma_\mu - \frac{\partial \Sigma(p)}{\partial p^\mu}$$

顶角 $T_\mu(k; p+k, p) = \gamma^\mu + \Lambda^\mu(k; p+k, p) \rightarrow \Lambda^\mu(0; p, p) = -\frac{\partial \Sigma(p)}{\partial p^\mu}$

6. 可重正性

1. Furry定理: 若封闭电子线有奇数个顶点, 其 $S=0$



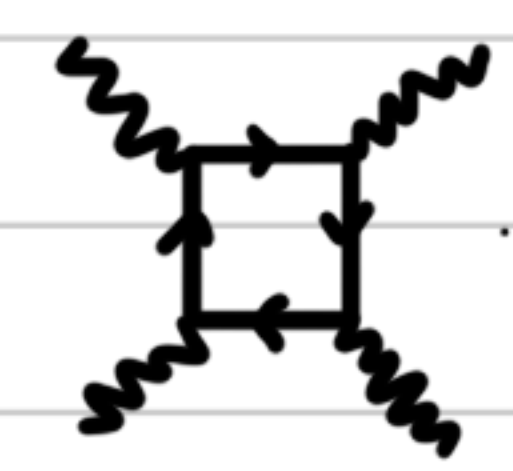
三光子顶角: S_a 含 $T_a = \text{tr}[\gamma^\lambda S_F(p) \gamma^\mu S_F(p) \gamma^\nu S_F(p)]$

反向: $C = i \gamma^2 \gamma^0 \rightarrow C^\dagger = i \gamma^0 \gamma^2$

$CC^\dagger = 1, C^\dagger \gamma_\mu C = -\tilde{\gamma}^\mu, C^\dagger S_F(p) C = \tilde{S}_F(-p) \rightarrow T_a = \text{tr}[C^\dagger \dots C] = (-1)^3 \text{tr}[\tilde{\gamma}^\lambda \tilde{S}_F(-p) \tilde{\gamma}^\mu \tilde{S}_F(-p) \tilde{\gamma}^\nu \tilde{S}_F(-p)]$

故 $S_a + S_b = 0$, 抵消

$= (-1)^3 T_b$



光子-光子散射 (无经典对应)

发散度 $D=0$

$$I_{\lambda\mu\nu\rho}(k_1, k_2, k_3, k_4) = \int d^4P R_{\lambda\mu\nu\rho}(k_1, k_2, k_3, k_4, P)$$

$$= I_{\lambda\mu\nu\rho}(0, 0, 0, 0, P) + R_{\lambda\mu\nu\rho}(k_1, k_2, k_3, k_4, P)$$

$$= I_{\lambda\mu\nu\rho}(0) + J_{\lambda\mu\nu\rho}(k_1, k_2, k_3, k_4) \leftarrow \text{单圈项}$$

$$S = I_{\lambda\mu\nu\rho}(0) \cdot A^\lambda A^\mu A^\nu A^\rho$$

无规范不变性, $I(0) = 0$

光子自能 $\Pi(k)$

$$D_{F\mu\nu}(k) = g_{\mu\nu} \frac{-1}{k^2 + i\epsilon}$$

$$D_{F\mu\nu}(k) = \frac{-g_{\mu\nu}}{k^2 + \Pi(k) + i\epsilon}$$

$$\text{完全传播子 } iD_{F\mu\nu}(k) = g_{\mu\nu} iD_F(k) \quad ; \quad i\Pi_{\mu\nu}(k) = g_{\mu\nu} i\Pi(k)$$

$$\text{展开: } iD_F(k) = iD_F(k) + iD_F(k)[-i\Pi(k)]iD_F(k) \dots$$

$$\leftarrow D_F(k)^{-1} = D_F(k)^{-1} - \Pi(k)$$

↓ 对 k^μ 导

$$\frac{\partial D_F(k)^{-1}}{\partial k^\mu} = -2k_\mu - \frac{\partial \Pi}{\partial k^\mu}$$

$$W_\mu = \Delta_\mu - 2k_\mu$$

可重正化:

骨架图形: 未修正的线与顶点 (最小不可约)

↓

既约图形

玻色子, Fermi 外线, Bose 子外线

判断: 表观发散度 $D = D(n, F, B)$ D 与 n 无关时可重正化

证明: